

AIKAWA, Japan

WMO: 476020

Lat: 38.0292N

Lon: 138.2400E

Elev: 24

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.0	31.2	18.4	13.9	35.4	20.0	15.0	35.9	43.4	35.5	40.5	36.0	20.0	320	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	88.6	77.0	86.4	76.7	84.4	76.0	79.0	85.1	78.1	84.1	77.0	83.1	8.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.8	83.3	76.2	136.9	82.4	75.1	131.6	81.4	42.6	85.4	41.6	84.2	40.5	83.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.7	32.6	28.7	27.2	93.9	1.9	3.1	25.8	96.2	24.7	98.0	23.7	99.7	22.3	101.9		
			24.9	80.3	1.5	1.2	23.8	81.2	22.9	82.0	22.1	82.7	21.0	83.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	38.8	38.7	43.2	51.8	60.7	67.7	75.5	79.0	72.5	62.8	52.8	44.0
	DBStd	14.70	4.36	4.83	5.47	5.96	5.39	4.14	4.60	3.95	4.95	4.93	5.61	5.31
	HDD50	1168	347	319	223	46	1	0	0	0	0	0	35	197
	HDD65	3915	811	737	677	397	153	20	0	0	3	101	366	650
	CDD50	3874	1	2	11	101	334	530	791	899	675	398	120	12
	CDD65	1142	0	0	0	1	21	100	326	434	227	33	0	0
	CDH74	7769	0	0	0	4	54	213	2160	4023	1260	55	0	0
	CDH80	1950	0	0	0	0	3	17	508	1164	254	4	0	0
(14) Wind	WSAvg	11.0	18.0	15.5	12.6	9.6	7.6	6.8	6.3	7.0	8.1	10.2	13.5	17.4
(15) Precipitation	PrecAvg	63.1	5.4	3.8	3.8	3.7	4.0	5.0	7.6	5.5	5.5	5.3	6.5	7.2
	PrecMax	81.1	10.7	7.7	7.1	6.5	9.2	9.3	18.7	15.7	13.7	9.1	11.2	15.2
	PrecMin	42.0	2.2	1.4	1.8	0.7	1.8	1.9	0.7	0.3	1.5	3.1	3.8	3.6
	PrecStd	9.4	1.7	1.4	1.2	1.6	2.0	2.5	4.6	3.3	3.0	1.7	1.6	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	58.3	63.6	72.9	79.1	82.2	91.5	93.3	89.6	78.9	69.0	61.6
		MCWB	48.6	49.5	50.7	58.0	63.0	69.7	76.3	77.1	76.7	68.6	60.1	54.6
	2%	DB	50.7	52.7	58.5	68.2	75.1	78.6	87.2	89.4	85.7	75.0	66.5	57.8
		MCWB	45.2	46.3	48.1	55.3	61.9	68.7	75.9	77.2	74.9	66.2	59.1	51.6
	5%	DB	48.3	49.4	55.2	64.8	72.3	76.3	85.0	87.5	82.9	72.8	64.5	55.3
		MCWB	43.2	44.3	47.8	53.7	60.9	67.9	76.0	77.6	74.0	64.7	57.5	49.5
	10%	DB	46.0	46.8	52.2	61.5	69.6	74.3	82.8	85.6	80.2	70.8	62.2	52.9
		MCWB	41.3	41.8	46.6	52.4	59.9	67.4	75.1	77.1	72.6	63.2	55.8	47.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	51.4	53.1	61.2	66.6	73.3	79.1	80.7	78.6	71.5	62.8	56.0
		MCDB	53.6	54.6	58.7	67.5	71.9	77.5	85.4	87.0	84.7	76.8	66.5	60.1
	2%	WB	46.1	47.4	50.8	57.5	64.5	71.4	77.9	79.6	76.3	68.2	60.6	52.3
		MCDB	50.0	51.8	55.9	64.6	71.0	75.1	84.2	85.8	83.0	73.2	65.0	56.8
	5%	WB	43.7	44.6	49.0	55.6	62.9	70.1	76.8	78.8	74.9	66.1	58.5	49.9
		MCDB	47.8	48.9	53.9	62.1	69.5	74.1	83.0	84.8	81.4	71.3	63.7	54.9
	10%	WB	41.6	42.1	47.0	53.8	61.2	68.7	75.8	78.0	73.5	64.1	56.3	47.8
		MCDB	45.7	46.6	52.0	59.6	67.6	73.0	81.6	84.0	79.8	69.6	61.5	52.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.8	10.7	12.3	12.4	10.5	9.1	10.2	10.7	11.2	10.6	8.9
		MCDBR	12.0	13.2	15.4	17.4	16.8	13.1	11.6	12.3	12.5	12.6	13.2	12.5
		MCWBR	10.5	11.5	11.4	10.2	9.2	7.4	6.2	6.8	7.2	8.5	10.8	11.0
	5% WB	MCDBR	11.7	12.8	13.5	14.9	14.3	10.0	10.2	10.5	11.1	11.0	12.3	12.3
		MCWBR	10.9	11.9	11.4	9.8	9.2	6.4	5.9	6.5	7.1	8.7	11.5	11.5
(40) Clear-Sky Solar Irradiance	taub		0.425	0.458	0.517	0.551	0.556	0.555	0.566	0.529	0.475	0.446	0.436	0.433
	taud		2.029	1.951	1.845	1.817	1.874	1.960	1.998	2.112	2.214	2.159	2.091	2.011
	Ebn at Noon		225	235	234	236	238	238	234	239	246	238	220	210
	Edh at Noon		40	49	61	66	63	58	56	49	42	40	38	39
(44) All-Sky Solar Radiation	RadAvg		454	713	1122	1496	1687	1681	1537	1617	1266	964	594	402
	RadStd		41	76	93	132	179	142	209	176	93	73	49	45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page